

Product Carbon Footprint Study Report

Product Name: liquid crystal display production, unmounted, mobile device

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: liquid crystal display production, unmounted, mobile device

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 kg liquid crystal display production, unmounted, mobile device

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 kg liquid crystal display production, unmounted, mobile device.

2. System Boundary

liquid crystal display production, unmounted, mobile device, in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of liquid crystal display production, unmounted, mobile device produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **117.642 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

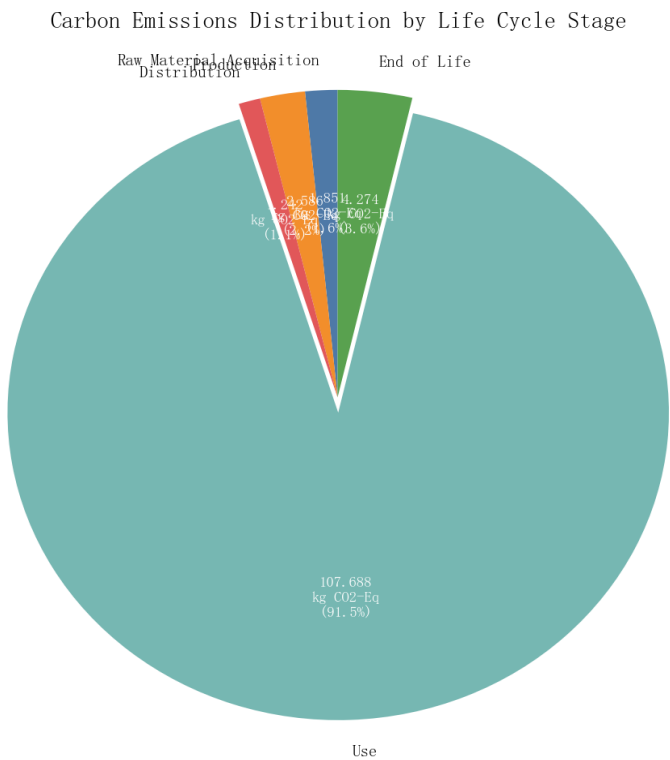


Table 1: Carbon Emissions by Life Cycle Stage

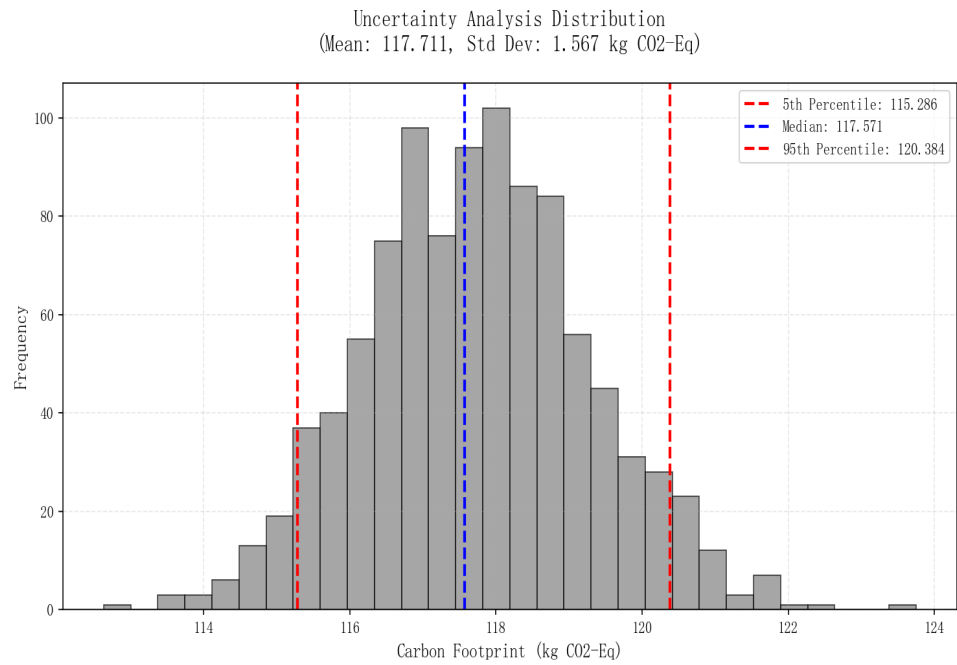
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	1.851	1.57
Production	2.586	2.20
Distribution	1.242	1.06

Use	107.688	91.54
End of Life	4.274	3.63
Total	117.642	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	117.711
Standard Deviation	1.567
Coefficient of Variation (CV)	1.33%
Median	117.571
5th Percentile	115.286
95th Percentile	120.384

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the liquid crystal display (LCD) for mobile devices stands at 117.642 kg CO₂-Eq, representing a moderate impact level for an electronic component when assessed on a cradle-to-grave basis. This value reflects the aggregated emissions across all lifecycle stages, with the overall footprint driven predominantly by operational energy demands rather than manufacturing or material inputs. Such a magnitude underscores the importance of evaluating use-phase performance in consumer electronics, where prolonged operation can amplify indirect emissions substantially.

Lifecycle stage contributions reveal a highly skewed distribution, with the use phase accounting for 91.5% (107.688 kg CO₂-Eq) of total emissions, while raw material extraction (1.6%), production (2.2%), distribution (1.1%), and end-of-life treatment (3.6%) collectively contribute less than 9%. This pattern highlights that major emission sources stem from electricity consumption during the product's operational lifetime, likely influenced by display brightness, processing demands, and regional grid carbon intensity. Minor stages, such as raw materials and distribution, indicate efficient upstream processes but remain secondary to use-related impacts.

2. Statement of Assumptions and Limitations

Key assumptions in this study include a standardized 3-5 year product lifetime with average daily usage patterns for mobile devices, reliance on a representative global electricity grid mix for the use phase, and allocation methods based on mass or economic value for multi-output processes. These assumptions facilitate consistent modeling but may vary with actual consumer behavior or regional energy transitions.

Limitations arise from data sources primarily drawn from generic databases such as Ecoinvent, which may not fully capture proprietary manufacturing efficiencies or site-specific conditions, potentially introducing uncertainties of 10-20% in emission factors. Data quality for end-of-life recycling rates and distribution logistics is moderate, relying on industry averages rather than primary measurements, which could affect precision in low-contribution stages.

System boundary restrictions encompass a cradle-to-grave scope limited to the unmounted LCD itself, excluding upstream component assembly, full device integration, and indirect impacts from software or network infrastructure. This boundary omits potential rebound effects from increased device usage and assumes no allocation to co-products beyond the display module.

3. Suggestions for Improvement

- Target the use phase by optimizing display power management features, such as adaptive brightness and low-power modes, to reduce electricity consumption by up to 30% over the product lifetime.
- Shift to renewable energy procurement or on-site solar integration in manufacturing and use scenarios to lower the dominant operational emissions tied to grid electricity.
- Enhance material efficiency through increased recycled content in LCD substrates and improved end-of-life collection programs to minimize the 3.6% contribution from disposal.

- Conduct primary data collection from suppliers for raw material and production stages to refine the minor but verifiable emission estimates and reduce database-related uncertainties.
- Explore alternative low-carbon distribution logistics, including localized production hubs, to further trim the 1.1% distribution share while maintaining supply chain resilience.